

What Do the Numbers Say? Reading an Evaluation Output

Module 2 · Day 1, Session 2: Descriptive Analysis, Live

3ie

1 • The case

The traffic camera programme

A roads authority is rolling out automated speed cameras. Twelve sectors got cameras in 2020; the other twelve were due in 2022.

Before extending the programme, it set a bar in the camera sectors: at least **2.0 fewer injury collisions per road segment between 2019 and 2021**. Below that, the money goes to other road safety work.

Programme	Automated speed enforcement
Rolled out	12 sectors in 2020, 12 in 2022
Outcome	Injury collisions
Decision rule	2.0 fewer per segment, 2019 to 2021

Illustrative case, with invented figures. This is not a real programme; we generated the numbers for this training. Every number in this session comes from this fictional dataset.

Session 1's three questions, on the full data

- How big?** In the units the decision is written in: collisions per segment
- How sure?** The p-value and the confidence interval
- Compared to what?** We come back to this at the end

We start from the raw data file, as an evaluator would, and build every number on screen.

You will not write any code. You call the numbers; I type and press **Run**. The grey boxes are real R, running in this page.

2 • Meet the data

One row, one road segment, one year

R Code

Start Over

Run Code

```
1 cols <- c("segment_id", "sector_id", "corridor_type", "year",
2           "treated", "injury_collisions")
3 head(pol[, cols], 4)
```

segment_id	sector_id	corridor_type	year	treated	injury_collisions
SEC01-001	SEC01	suburban arterial	2019	1	10
SEC01-001	SEC01	suburban arterial	2021	1	9
SEC01-002	SEC01	ring road	2019	1	4
SEC01-002	SEC01	ring road	2021	1	2

treated is 1 if the segment's sector got cameras in 2020. **injury_collisions** is the number of collisions with an injury on that segment in that year.

The file has 2,400 rows. How many **road segments** is that?

1,200: each segment appears twice, once in 2019 and once in 2021.

Who is in the data?

R Code

↺ Start Over

▶ Run Code

```
1 base <- subset(pol, year == 2019)
2 data.frame(
3   segments = tapply(base$segment_id, base$treated, length),
4   sectors  = tapply(base$sector_id, base$treated, function(x) length(unique(x))))
```

	segments	sectors
0	600	12
1	600	12

Row **1** is the sectors that got cameras in 2020; row **0** is the sectors still waiting. Half the road network in each.

Before we look at any collisions: what would you want to know about **how** the first twelve sectors were chosen?

Word 1 · The average

R Code

↺ Start Over

▶ Run Code

```
1 summary(pol$injury_collisions[pol$year == 2019])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	6.00	9.00	10.44	13.00	42.00

`summary()` gives six numbers for one column, here 2019: the smallest, the largest, the middle road (**median**), the **mean**, and the quartiles (a quarter of roads sit below **1st Qu.**, a quarter above **3rd Qu.**).

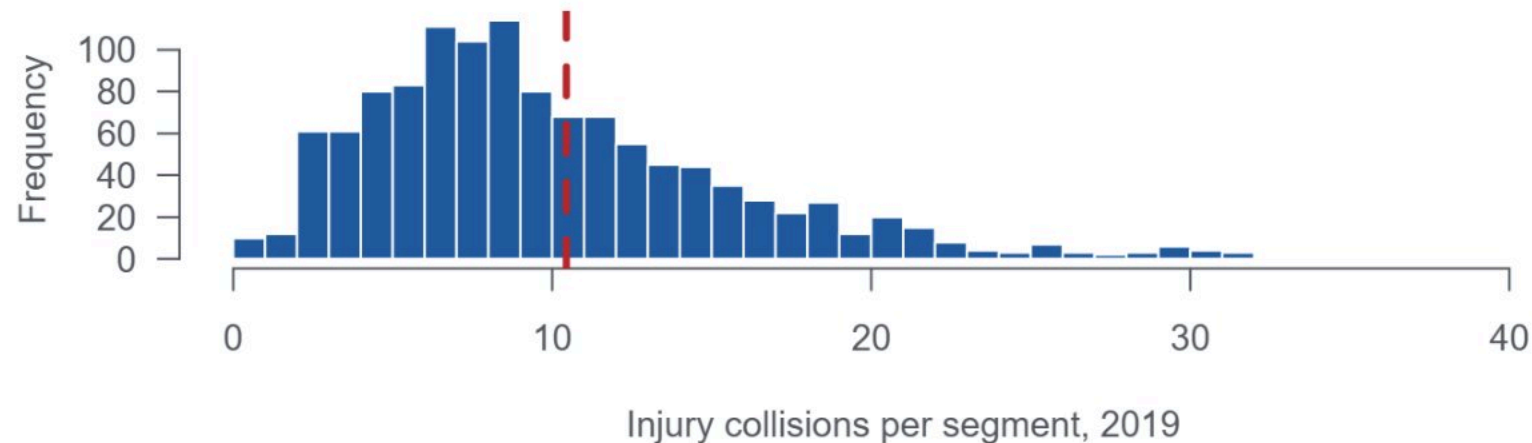
The mean is about 10.4 collisions. Is a typical road segment close to 10.4? What do the minimum and maximum suggest?

Word 1 · What the average hides

R Code [Start Over](#)

[Run Code](#)

```
1 before <- subset(pol, year == 2019)
2 clean_par()
3 hist(before$injury_collisions, breaks = 40, col = ACCENT, border = "white",
4       main = "", xlab = "Injury collisions per segment, 2019")
5 abline(v = mean(before$injury_collisions), col = ALERT, lwd = 3, lty = 2)
```



About seven in ten segments had 5 to 15 collisions in 2019, and nine had more than 30. Those few pull the mean (red line) above the median. In Session 1, one busy ring road pulled the mean of ten roads to 13.5; the median stayed at 10.

3 · Graph it

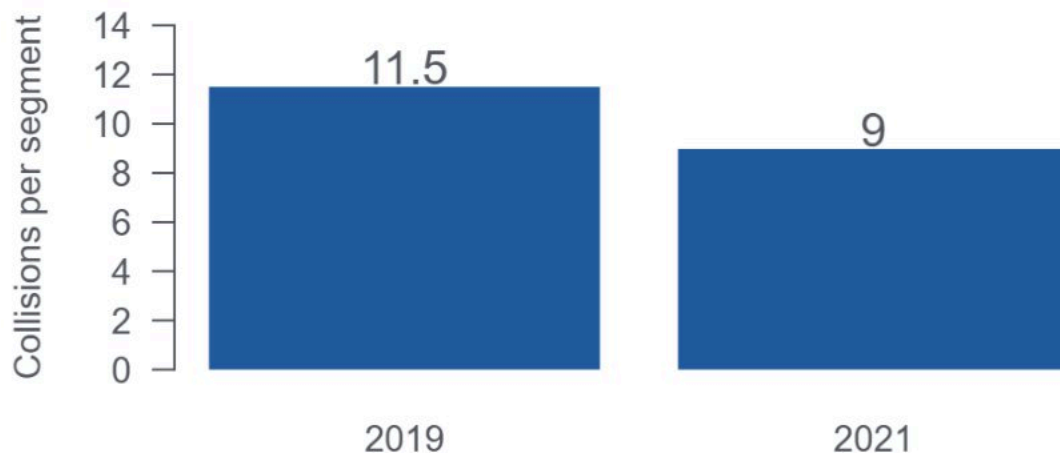
Before and after, where cameras went

R Code

↺ Start Over

▶ Run Code

```
1 cams <- subset(pol, treated == 1)
2 means <- tapply(cams$injury_collisions, cams$year, mean)
3 clean_par()
4 b <- barplot(means, col = ACCENT, border = NA, ylim = c(0, 14),
5             ylab = "Collisions per segment")
6 text(b, means + 0.8, round(means, 1), cex = 1.3)
```



This is the chart a press release would use. What does it show, and what does it not show?

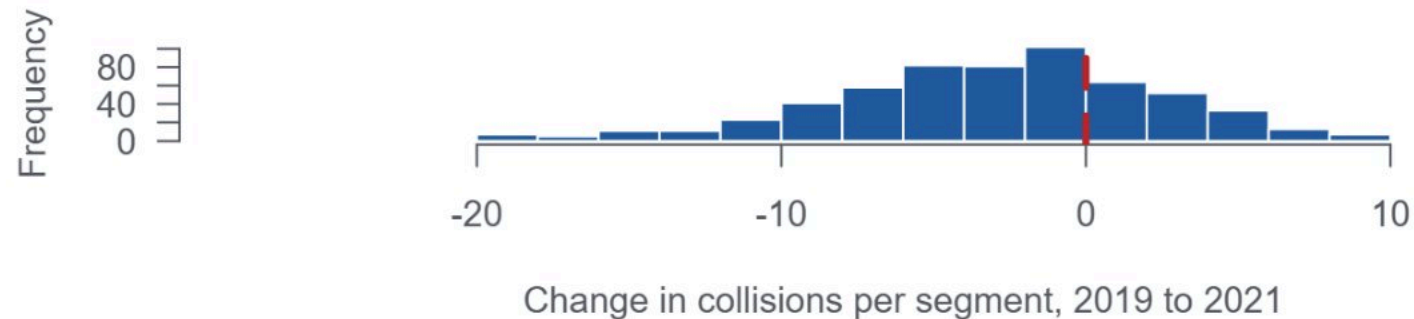
Did every road fall?

R Code

Start Over

Run Code

```
1 wide <- reshape(cams[, c("segment_id", "year", "injury_collisions")],
2                 idvar = "segment_id", timevar = "year", direction = "wide")
3 change <- wide$injury_collisions.2021 - wide$injury_collisions.2019
4 clean_par()
5 hist(change, breaks = 30, col = ACCENT, border = "white", main = "",
6      xlab = "Change in collisions per segment, 2019 to 2021")
7 abline(v = 0, col = ALERT, lwd = 3, lty = 2)
```



Before I run it: out of ten camera roads, how many had fewer collisions in 2021?

About six in ten (63%); the rest stayed the same or went up. Word 1 again: an average change hides how much roads differ.

Would you fund the extension?

Collisions fell from 11.5 to 9.0 per segment where cameras went. **Would you commit the money to extend them?**

Yes · No · Not on this alone

Go to **menti.com** and enter code **1234 5678**, or scan the code on the right.



4 • How big, and how sure?

How big? Work it out from the averages

R Code

↺ Start Over

▶ Run Code

```
1 before <- mean(cams$injury_collisions[cams$year == 2019])
2 after  <- mean(cams$injury_collisions[cams$year == 2021])
3 round(data.frame(before, after, change = after - before,
4                   percent = 100 * (after - before) / before), 2)
```

before	after	change	percent
11.5	8.97	-2.53	-21.96

The rule asks for **2.0 fewer collisions per segment**. Which of these numbers do you check against it?

The change, -2.53, is in the rule's units; the 22% is not.

Word 3 · The regression gives the same number

R Code

↻ Start Over

▶ Run Code

```
1 fit <- lm(injury_collisions ~ round, data = cams)
2 coef(fit)
```

(Intercept)	round
11.49833	-2.52500

round is 0 for 2019 and 1 for 2021. The table an evaluator hands you comes from a line like this one.

Which number is the 2019 average, and which is the change?

(Intercept) is the 2019 average, 11.50. **round** is the change, -2.53: the same number we got by subtracting two averages. Session 1's Word 3: a coefficient is a gap between two groups. Here the two groups are the same roads in two years.

Words 4 and 5 · How sure?

R Code [Start Over](#)

[Run Code](#)

```
1 fit <- lm_robust(injury_collisions ~ round, data = cams, clusters = segment_id)
2 round(summary(fit)$coefficients[, c(1, 2, 4, 5, 6)], 3)
```

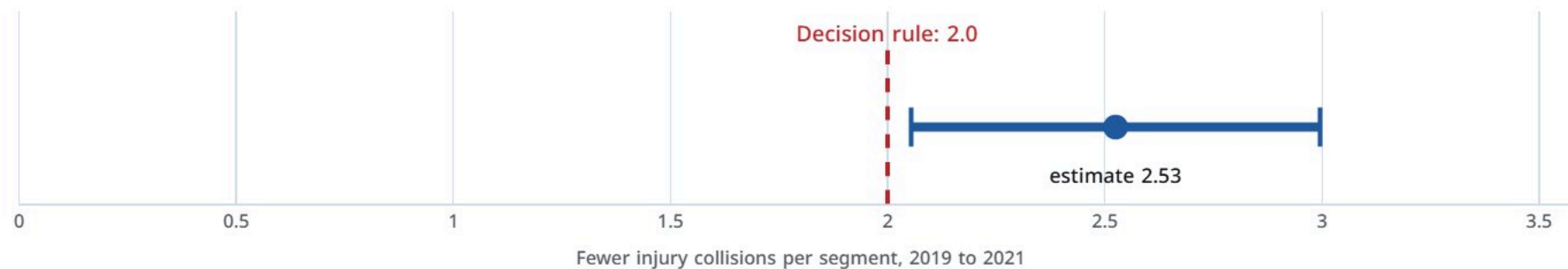
	Estimate	Std. Error	Pr(> t)	CI Lower	CI Upper
(Intercept)	11.498	0.259	0	10.990	12.007
round	-2.525	0.240	0	-2.996	-2.054

Before I press Run: will the whole interval sit above 2.0 fewer, or will it cross 2.0?

Pr(>|t|) is the p-value. Printed as 0, it means below 0.001: a fall this size would be very unlikely if nothing had changed.

CI Lower and **CI Upper** are the 95% confidence interval: the fall is somewhere between 2.05 and 3.00 collisions per segment.

Does it clear the rule?



Decision rule	2.00 fewer
The change	2.53 fewer
95% confidence interval	2.05 to 3.00

The estimate clears the rule, and so does the whole interval, just. On **how big** and **how sure**, the programme passes.

Which question does this chart leave open?

5 • Put AI to work

AI Snapshot: ask AI to read the chart

Give a picture of the before-and-after bar chart to an AI tool with the prompt most people would type: **“What does this chart show? Did the cameras work?”**

In pairs: which of these claims can you check against the chart, and which does the chart support?

Illustrative response. A response like this is possible, especially with a short or leading prompt:

“The chart shows a 22% reduction in injury collisions after speed cameras were installed, from 11.5 to 9.0 per road segment. This reduction is statistically significant ($p < 0.001$), confirming that the cameras caused the decline. The fall was consistent across all road types, which suggests the effect is robust. Given the size of the effect, the programme should be extended to the remaining sectors.”

Debrief: what the AI added

"22% reduction, 11.5 to 9.0"

Correct. The 22% can be worked out from the two bars.

" $p < 0.001$ "

Not on the chart. The model supplied a p-value it was never given.

"confirming that the cameras caused"

A before-and-after chart cannot show cause. A small p-value says the fall is unlikely to be noise, not what caused it.

"consistent across all road types"

Invented. The chart has no road types, and more than a third of camera roads did not fall.

"should be extended"

A recommendation with no rule and no comparison.

Only the first sentence comes from the chart. Everything after it is the model filling gaps with what often follows a chart like this. Check whether yours did the same.

A better prompt

“This chart shows average injury collisions per road segment in the sectors that got speed cameras, in 2019 and 2021. Describe only what the chart shows. Then list what else could explain the change, and what comparison you would need before saying the cameras caused it. Before you answer, ask me any questions you need.”

Run both prompts on the same chart. What does the second answer include that the first left out? Paste the second answer into a fresh chat: which claims does it flag that the chart cannot support?

Two habits for every session: ask the AI to **ask you questions first**, and have a second chat review what the first one wrote.

6 · Compared to what?

What else happened between 2019 and 2021?

So far we have compared the camera roads with **their own past**. The fall of 2.53 includes anything else that changed over those two years.

At your table: what two or three things apart from cameras could have cut collisions in these sectors? What would you compare the camera roads with to rule them out?

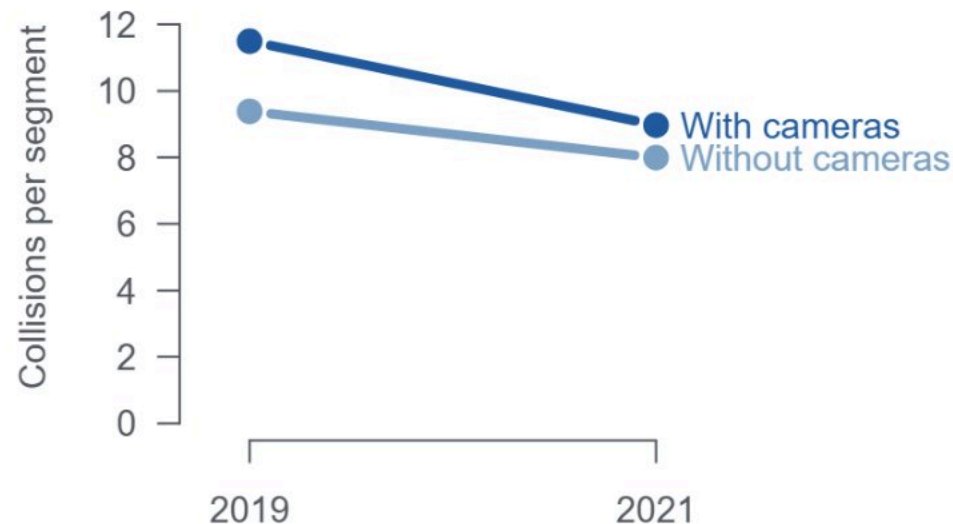
One comparison: roads without cameras

R Code

↺ Start Over

▶ Run Code

```
1 means <- tapply(pol$injury_collisions,  
2               list(pol$year, pol$treated), mean)  
3 plot_two_groups(means)
```



In the twelve sectors still waiting for cameras, which way did collisions go?

Down too, by 1.39 per segment (15%), with no cameras installed. Some of the 2.53 would have happened anyway.

How much, and whether these roads are a fair comparison, is the subject of Session 3.

Look back at your vote

[PLACEHOLDER] Live Menti results. Set [MENTI_EMBED](#) in the setup chunk to show the room vote here.

You voted on two before-and-after averages. They cannot answer “compared to what?”.

Who would change their vote now? Can we hear one reason from someone who changed, and one from someone who did not?

7 • Take this back to your desk

Annotate this output

	Estimate	Std. Error	p-value	CI Lower	CI Upper
(Intercept)	11.498	0.259	<0.001	10.990	12.007
round	-2.525	0.240	<0.001	-2.996	-2.054

Outcome: injury collisions per road segment, camera sectors only, 2019 (round 0) and 2021 (round 1). Standard errors clustered by road segment. The p-value is the column R printed as $\text{Pr}(> |t|)$.

On your printed copy:

1. Which number is the before-and-after change? Circle it.
2. Where is its confidence interval? Draw a box around it.
3. Which number is the 2019 average? Underline it. Why is it not an effect?
4. What comparison would you ask the evaluator for?

Three questions to ask the evaluator

- How big?** In the units the decision is written in, not a percentage
- How sure?** Does the confidence interval clear the rule, not just zero?
- Compared to what?** Which group, which period, and did it move too?

The camera result passes the first two questions and clears the rule. The third is still open, and Session 3 takes it up.

The last figure you were sent: was it before-and-after? What would you compare it with?